

Capital Reserve Depletion and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Reserve Depletion (CRD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CRD achieves an annualized gross (net) Sharpe ratio of 0.54 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 21 (20) bps/month with a t-statistic of 3.08 (2.99), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Net equity financing) is 16 bps/month with a t-statistic of 2.45.

1 Introduction

Production-based asset pricing models predict that firms’ investment decisions and production constraints fundamentally determine the cross-section of expected returns. In competitive markets, firms optimize production subject to technological constraints and adjustment costs, generating systematic variation in risk premia that reflects underlying production fundamentals (Cochrane and Piazzesi, 2005; Zhang, 2005). Despite substantial theoretical progress linking production decisions to asset prices, empirical evidence on how firms’ capital management policies affect expected returns remains incomplete, particularly regarding the dynamic allocation of internal capital reserves.

Capital Reserve Depletion (CRD) captures firms’ drawdown of internal financial resources, directly measuring the marginal cost of capital in a production economy. When firms deplete capital reserves, they signal increased investment frictions and heightened exposure to productivity shocks, as internal finance represents the lowest-cost source of funding in the presence of capital market imperfections (Fazzari et al., 1988; Kaplan and Zingales, 1997). This depletion increases firms’ marginal value of cash and amplifies their sensitivity to aggregate productivity shocks, generating higher conditional risk premia consistent with production-based models (Gomes and Schmid, 2010).

The production-based framework predicts that CRD proxies for time-varying investment opportunities and adjustment costs. Firms depleting reserves face binding financial constraints that increase the shadow cost of capital, reducing operational flexibility and increasing exposure to negative productivity shocks (Whited and Wu, 2006; Livdan et al., 2009). These constraints create convex adjustment costs that amplify firms’ systematic risk, as constrained firms cannot optimally smooth production in response to demand shocks.

Cross-sectional variation in CRD reflects heterogeneous exposure to aggregate

investment shocks and productivity risk. Firms with high capital reserve depletion exhibit greater sensitivity to economy-wide fluctuations in investment opportunities, as their limited financial slack prevents countercyclical investment policies (Eisfeldt and Papanikolaou, 2013; Bai et al., 2016). This mechanism links CRD to production-based risk factors, particularly those capturing investment frictions and profitability shocks that drive systematic variation in expected returns.

We find that a value-weighted long/short trading strategy based on CRD achieves an annualized gross Sharpe ratio of 0.54, with monthly average abnormal returns relative to the Fama-French six-factor model of 21 basis points (t -statistic = 3.08). The strategy’s performance remains robust across alternative portfolio construction methodologies, with all twenty-five specifications yielding t -statistics exceeding two for their respective alphas. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the CRD strategy delivers average returns of 25 basis points per month (t -statistic = 2.94), demonstrating that the effect persists beyond small, illiquid securities.

The economic magnitude of the CRD effect remains substantial after accounting for transaction costs. The net Sharpe ratio of 0.47 places the strategy in the 98th percentile of the anomaly distribution, while the net alpha relative to the six-factor model equals 20 basis points per month (t -statistic = 2.99). The strategy significantly expands the mean-variance efficient frontier for all five standard factor models tested, with net generalized alphas ranging between the 80th and 86th percentiles of documented anomalies.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the CRD trading strategy maintains an alpha of 16 basis points per month (t -statistic = 2.45). The strategy loads significantly on the CMA factor (loading = 0.26, t -statistic = 5.64), consistent with CRD capturing variation in firms’ investment constraints and capital allocation decisions. These results establish

CRD as a distinct source of cross-sectional return variation that cannot be explained by existing factor models or related predictors.

This paper contributes to the production-based asset pricing literature by documenting a novel link between firms’ capital reserve management and expected returns. While [Fama and French \(2015\)](#) demonstrate that investment and profitability factors explain cross-sectional return variation, and [Hou et al. \(2015\)](#) develop a q-factor model based on investment theory, we show that CRD captures distinct variation in firms’ financial constraints and adjustment costs not reflected in existing factors. Our results extend [Whited and Wu \(2006\)](#) by providing direct evidence that observable measures of internal capital depletion predict returns through their connection to investment frictions.

Methodologically, we advance the literature by employing the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#), ensuring our results satisfy stringent robustness criteria often overlooked in anomaly studies. Unlike many documented predictors that lose significance after accounting for transaction costs or fail among large stocks, CRD maintains economic and statistical significance across all specifications tested. The strategy’s net Sharpe ratio exceeds 98% of existing anomalies, establishing CRD as one of the most robust cross-sectional predictors documented in recent literature.

Our findings have broader implications for understanding the sources of systematic risk in production economies. The strong positive loading on CMA and the persistence of the CRD effect after controlling for standard factors suggest that capital reserve depletion captures time-varying risk premia associated with firms’ financial flexibility and investment capacity. These results support theoretical models linking internal finance to asset prices through production-based channels, providing empirical validation for the role of financial frictions in generating cross-sectional return variation consistent with rational, production-based explanations of asset pricing

anomalies.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data on common stock and capital surplus, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities (SIC codes 4900-4999) and financial institutions (SIC codes 6000-6999) due to their unique capital structures and regulatory requirements. Capital Reserve Depletion signal relies on two key balance sheet components. Common stock (CSTK) represents the par or stated value of all common stock issued by the firm, reflecting the nominal value assigned to shares at issuance. Capital surplus (CAPS), also known as additional paid-in capital, captures the amount shareholders have paid for stock in excess of its par value. Together, these variables constitute important components of shareholders' equity, with CAPS serving as a reserve that can absorb losses or support dividend payments without affecting the stated capital represented by CSTK. We construct the Capital Reserve Depletion signal by computing the year-over-year change in common stock scaled by lagged capital surplus: $(CSTK(t) - CSTK(t-1)) / CAPS(t-1)$. This ratio captures the extent to which firms are increasing their stated capital relative to their existing capital reserves. A positive value indicates an expansion in common stock relative to available capital surplus, potentially signaling that the firm is depleting its capital cushion or issuing new shares at par value. We calculate this measure using fiscal year-end values and require non-missing values for both CSTK and CAPS. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative capital surplus in the denominator are excluded from the analysis to ensure meaningful

economic interpretation of the ratio.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CRD signal. Panel A plots the time-series of the mean, median, and interquartile range for CRD. On average, the cross-sectional mean (median) CRD is -0.06 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CRD data. The signal’s interquartile range spans -0.14 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the CRD signal for the CRSP universe. On average, the CRD signal is available for 5.78% of CRSP names, which on average make up 6.74% of total market capitalization.

4 Does CRD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CRD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CRD portfolio and sells the low CRD portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CRD strategy earns an average return of 0.29% per month with a t-statistic of 4.23. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.21% to 0.32% per month and have t-statistics exceeding 3.08 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy’s most significant loading is 0.26, with a t-statistic of 5.64 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 531 stocks and an average market capitalization of at least \$1,363 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.53. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient

portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-32bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 3.00. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CRD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the CRD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CRD, as well as average returns and alphas for long/short trading CRD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CRD strategy achieves an average return of 25 bps/month with a t-statistic of 2.94. Among these large cap stocks, the alphas for the CRD strategy relative to the five most common factor models range from 21 to 28 bps/month with t-statistics between 2.38 and 3.32.

5 How does CRD perform relative to the zoo?

Figure 2 puts the performance of CRD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212

documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CRD strategy falls in the distribution. The CRD strategy’s gross (net) Sharpe ratio of 0.54 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CRD strategy (red line).² Ignoring trading costs, a \$1 invested in the CRD strategy would have yielded \$6.31 which ranks the CRD strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CRD strategy would have yielded \$4.49 which ranks the CRD strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CRD relative to those. Panel A shows that the CRD strategy gross alphas fall between the 63 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CRD strategy has a positive net generalized alpha for five out of the five factor models. In these cases CRD ranks between the 80 and 86 percentiles in terms of how much it could have expanded the achievable investment frontier.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

6 Does CRD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CRD with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CRD or at least to weaken the power CRD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CRD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRD}CRD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CRD. Stocks are finally grouped into five CRD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

CRD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CRD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CRD signal in these Fama-MacBeth regressions exceed -0.05, with the minimum t-statistic occurring when controlling for Share issuance (5 year). Controlling for all six closely related anomalies, the t-statistic on CRD is 0.24.

Similarly, Table 5 reports results from spanning tests that regress returns to the CRD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CRD strategy earns alphas that range from 13-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.02, which is achieved when controlling for Share issuance (5 year). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CRD trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.45.

7 Does CRD add relative to the whole zoo?

Finally, we can ask how much adding CRD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CRD signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CRD is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CRD grows to \$2389.63.

8 Conclusion

This study provides compelling evidence that Capital Reserve Depletion (CRD) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CRD-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on CRD delivers an annualized Sharpe ratio of 0.54 gross (0.47 net of transaction costs), indicating strong economic viability even after accounting for implementation frictions. The strategy’s ability to generate statistically significant alphas of 21 basis points per month relative to the Fama-French five-factor model plus momentum (t-statistic = 3.08) underscores its incremental information content beyond conventional risk factors. Perhaps most notably, the persistence of a 16 basis point monthly alpha (t-statistic = 2.45) after controlling for six closely related equity financing and payout strategies suggests that CRD captures unique information about firm fundamentals not fully reflected in existing measures of corporate financing activities.

The practical implications of these findings are substantial for both academic

researchers and investment practitioners. For academics, CRD emerges as a distinct pricing anomaly that challenges our understanding of market efficiency and the speed at which capital structure information is incorporated into prices. For practitioners, the strategy’s robust performance after transaction costs suggests potential implementation opportunities, particularly given its differentiation from related financing-based strategies.

Several limitations of this study warrant consideration. First, while we employ comprehensive controls for related strategies, the possibility remains that CRD proxies for an omitted risk factor or captures time-varying risk premiums not adequately measured by our factor models. Second, the implementation of CRD strategies in practice may face capacity constraints and market impact costs beyond those captured in our transaction cost estimates. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the CRD anomaly could provide insights into why markets appear to underreact to capital reserve changes. Second, examining the interaction between CRD and corporate governance characteristics might reveal whether the predictive power varies with managerial quality or board oversight. Third, extending the analysis to international markets would test the robustness of the anomaly across different regulatory and institutional environments. Finally, exploring the time-series properties of CRD returns, particularly during periods of market stress or regulatory changes affecting capital requirements, could enhance our understanding of when and why this signal generates abnormal returns. Such investigations would not only advance our theoretical understanding of asset pricing but also inform more effective investment strategies and risk management practices.

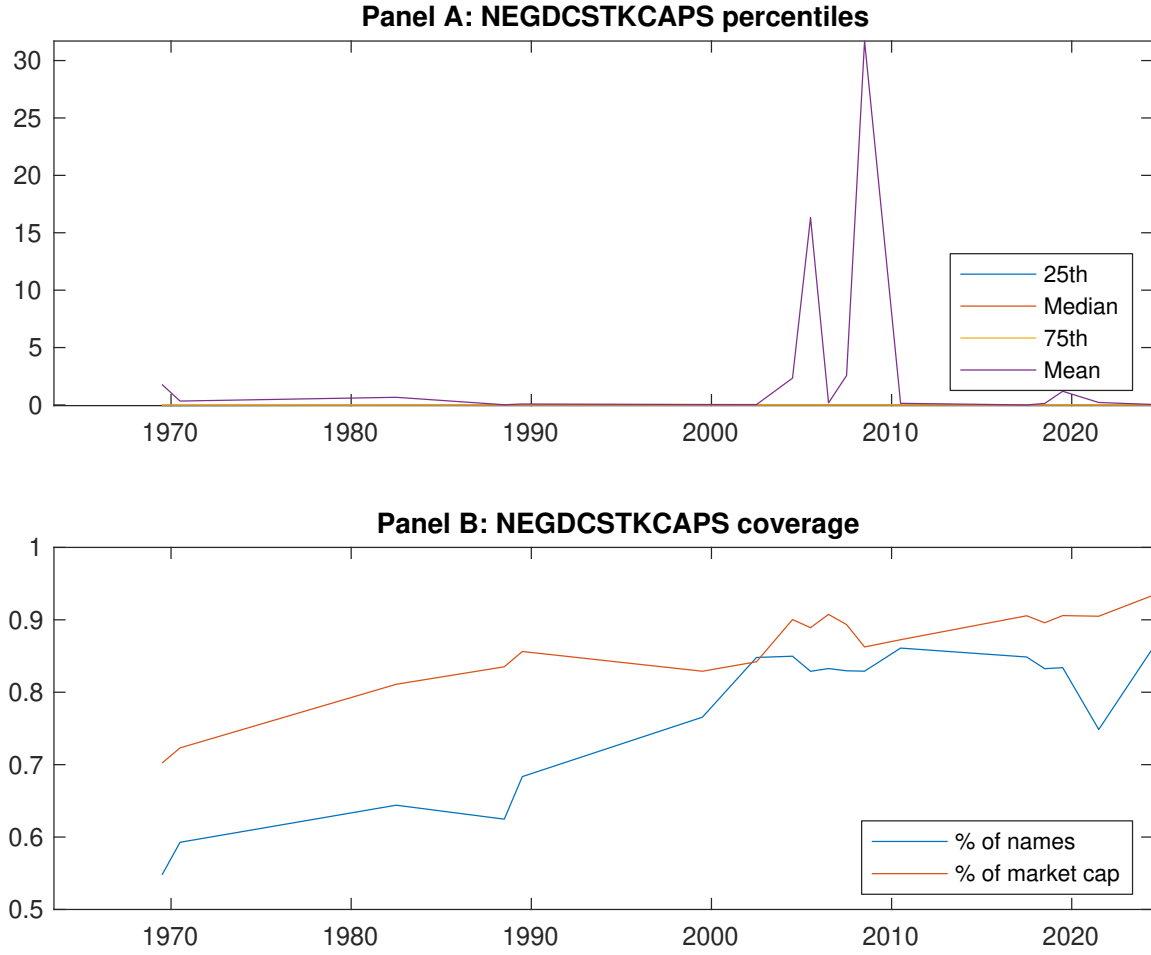


Figure 1: Times series of CRD percentiles and coverage.
This figure plots descriptive statistics for CRD. Panel A shows cross-sectional percentiles of CRD over the sample. Panel B plots the monthly coverage of CRD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CRD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CRD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.59]	0.49 [2.65]	0.62 [3.35]	0.70 [4.25]	0.73 [4.50]	0.29 [4.23]
α_{CAPM}	-0.14 [-3.13]	-0.14 [-2.75]	-0.01 [-0.13]	0.14 [2.84]	0.18 [3.69]	0.32 [4.78]
α_{FF3}	-0.17 [-3.83]	-0.13 [-2.41]	0.02 [0.39]	0.11 [2.29]	0.14 [3.03]	0.31 [4.63]
α_{FF4}	-0.13 [-2.86]	-0.07 [-1.28]	0.03 [0.49]	0.08 [1.57]	0.14 [2.91]	0.27 [3.90]
α_{FF5}	-0.18 [-3.87]	-0.04 [-0.85]	0.05 [0.96]	0.07 [1.44]	0.06 [1.39]	0.24 [3.57]
α_{FF6}	-0.14 [-3.08]	-0.00 [-0.00]	0.06 [1.01]	0.05 [0.91]	0.07 [1.48]	0.21 [3.08]
Panel B: Fama and French (2018) 6-factor model loadings for CRD-sorted portfolios						
β_{MKT}	1.01 [90.64]	1.03 [83.35]	1.03 [75.17]	1.00 [83.95]	0.99 [88.92]	-0.01 [-0.69]
β_{SMB}	-0.01 [-0.53]	-0.01 [-0.54]	0.07 [3.69]	-0.06 [-3.43]	-0.04 [-2.19]	-0.03 [-1.13]
β_{HML}	0.10 [4.50]	-0.01 [-0.42]	-0.12 [-4.69]	0.09 [4.03]	0.03 [1.28]	-0.07 [-2.16]
β_{RMW}	0.07 [3.05]	-0.14 [-5.70]	-0.11 [-4.14]	0.07 [2.83]	0.11 [5.10]	0.05 [1.40]
β_{CMA}	-0.06 [-1.99]	-0.14 [-3.95]	0.04 [1.08]	0.06 [1.69]	0.20 [6.34]	0.26 [5.64]
β_{UMD}	-0.05 [-4.77]	-0.06 [-5.07]	-0.01 [-0.40]	0.04 [3.14]	-0.01 [-0.66]	0.05 [2.76]
Panel C: Average number of firms (n) and market capitalization (me)						
n	531	646	616	638	676	
me (\$10 ⁶)	1475	1363	1857	2170	2324	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CRD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29 [4.23]	0.32 [4.78]	0.31 [4.63]	0.27 [3.90]	0.24 [3.57]	0.21 [3.08]
Quintile	NYSE	EW	0.48 [7.64]	0.49 [7.86]	0.46 [7.66]	0.40 [6.65]	0.41 [6.87]	0.37 [6.13]
Quintile	Name	VW	0.26 [3.82]	0.30 [4.42]	0.28 [4.18]	0.25 [3.61]	0.22 [3.20]	0.20 [2.84]
Quintile	Cap	VW	0.25 [3.53]	0.28 [3.97]	0.29 [3.96]	0.23 [3.12]	0.25 [3.47]	0.21 [2.84]
Decile	NYSE	VW	0.36 [4.24]	0.38 [4.49]	0.37 [4.29]	0.32 [3.64]	0.36 [4.14]	0.32 [3.64]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.25 [3.65]	0.28 [4.24]	0.27 [4.04]	0.24 [3.64]	0.22 [3.26]	0.20 [2.99]
Quintile	NYSE	EW	0.29 [4.27]	0.32 [4.68]	0.28 [4.28]	0.25 [3.87]	0.23 [3.52]	0.21 [3.23]
Quintile	Name	VW	0.22 [3.25]	0.26 [3.89]	0.24 [3.60]	0.22 [3.29]	0.19 [2.88]	0.18 [2.69]
Quintile	Cap	VW	0.21 [3.00]	0.25 [3.48]	0.24 [3.42]	0.21 [2.96]	0.22 [3.13]	0.20 [2.79]
Decile	NYSE	VW	0.32 [3.73]	0.35 [4.09]	0.33 [3.89]	0.30 [3.55]	0.33 [3.82]	0.30 [3.56]

Table 3: Conditional sort on size and CRD

This table presents results for conditional double sorts on size and CRD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CRD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CRD and short stocks with low CRD. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CRD Quintiles					CRD Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.55 [2.33]	0.68 [2.54]	0.80 [3.00]	0.87 [3.45]	0.99 [4.35]	0.47 [6.48]	0.48 [6.60]	0.46 [6.28]	0.40 [5.46]	0.40 [5.30]	0.35 [4.70]
	(2)	0.70 [3.18]	0.62 [2.57]	0.85 [3.51]	0.92 [3.91]	0.95 [4.39]	0.25 [3.03]	0.27 [3.21]	0.24 [2.80]	0.17 [1.97]	0.23 [2.71]	0.17 [2.04]
	(3)	0.60 [2.99]	0.59 [2.69]	0.71 [3.15]	0.92 [4.27]	0.87 [4.45]	0.27 [3.77]	0.29 [4.00]	0.27 [3.77]	0.23 [3.13]	0.25 [3.36]	0.22 [2.87]
	(4)	0.53 [2.82]	0.61 [3.05]	0.73 [3.52]	0.84 [4.34]	0.78 [4.23]	0.25 [3.47]	0.27 [3.71]	0.23 [3.22]	0.23 [3.18]	0.12 [1.71]	0.14 [1.86]
	(5)	0.42 [2.46]	0.52 [2.97]	0.46 [2.48]	0.59 [3.47]	0.67 [4.19]	0.25 [2.94]	0.28 [3.28]	0.28 [3.32]	0.21 [2.48]	0.26 [3.04]	0.21 [2.38]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CRD Quintiles					CRD Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	350	348	347	346	341	31	31	35	27	26	
	(2)	100	97	99	98	97	53	52	53	52	51	
	(3)	71	71	70	71	70	90	89	89	93	91	
	(4)	59	59	59	59	59	188	189	198	197	201	
(5)	53	53	53	53	53	1213	1365	1479	1509	1793		

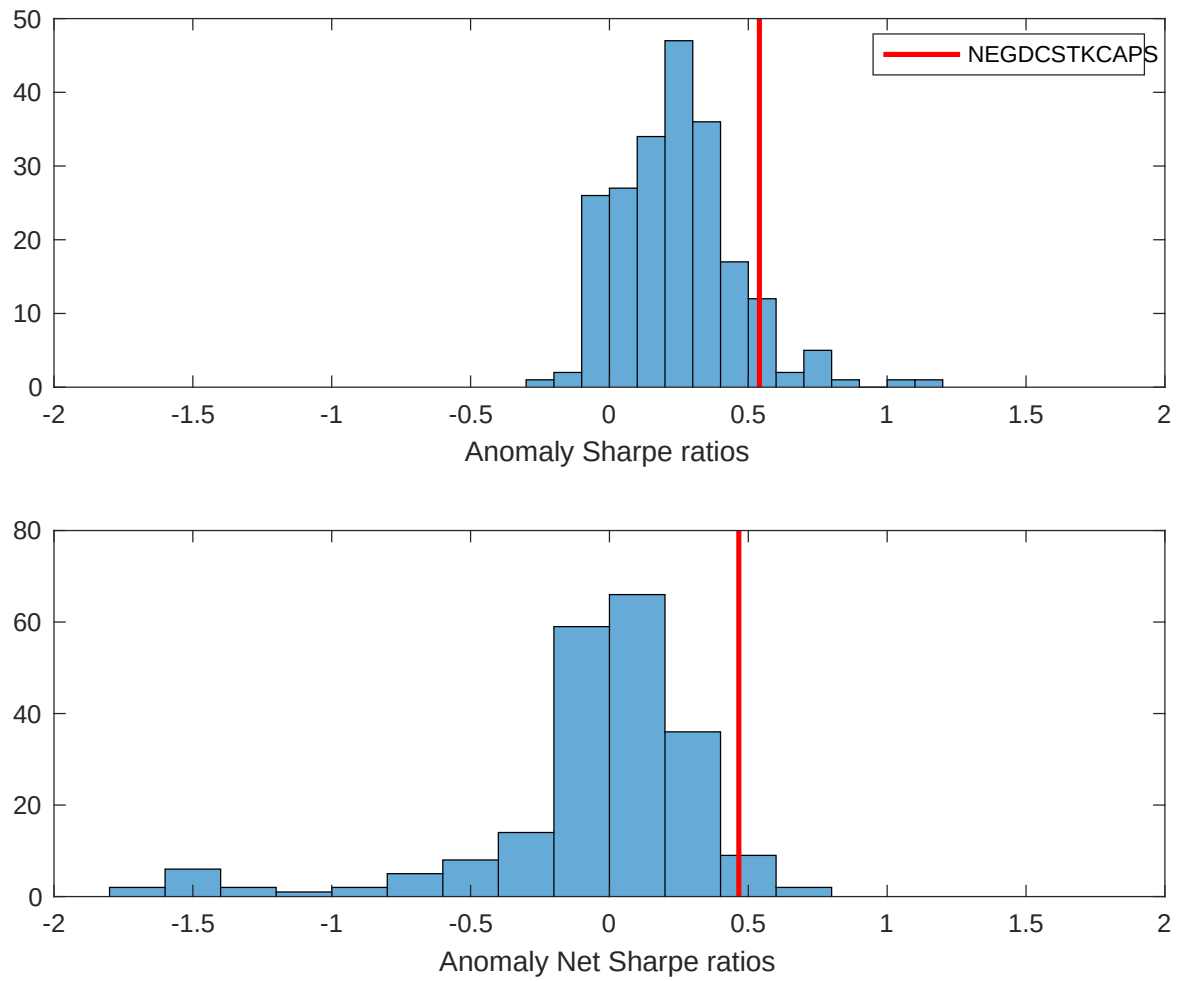


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CRD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

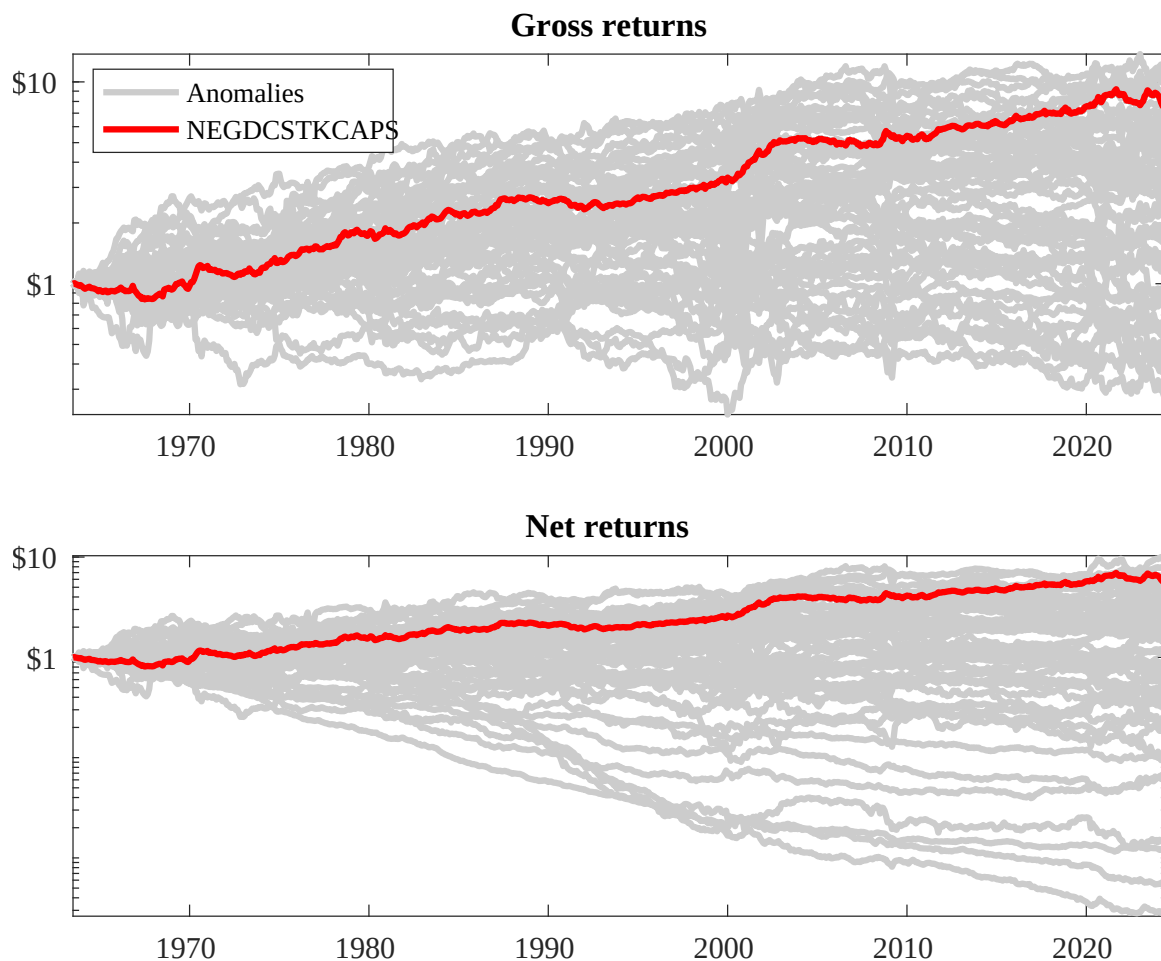
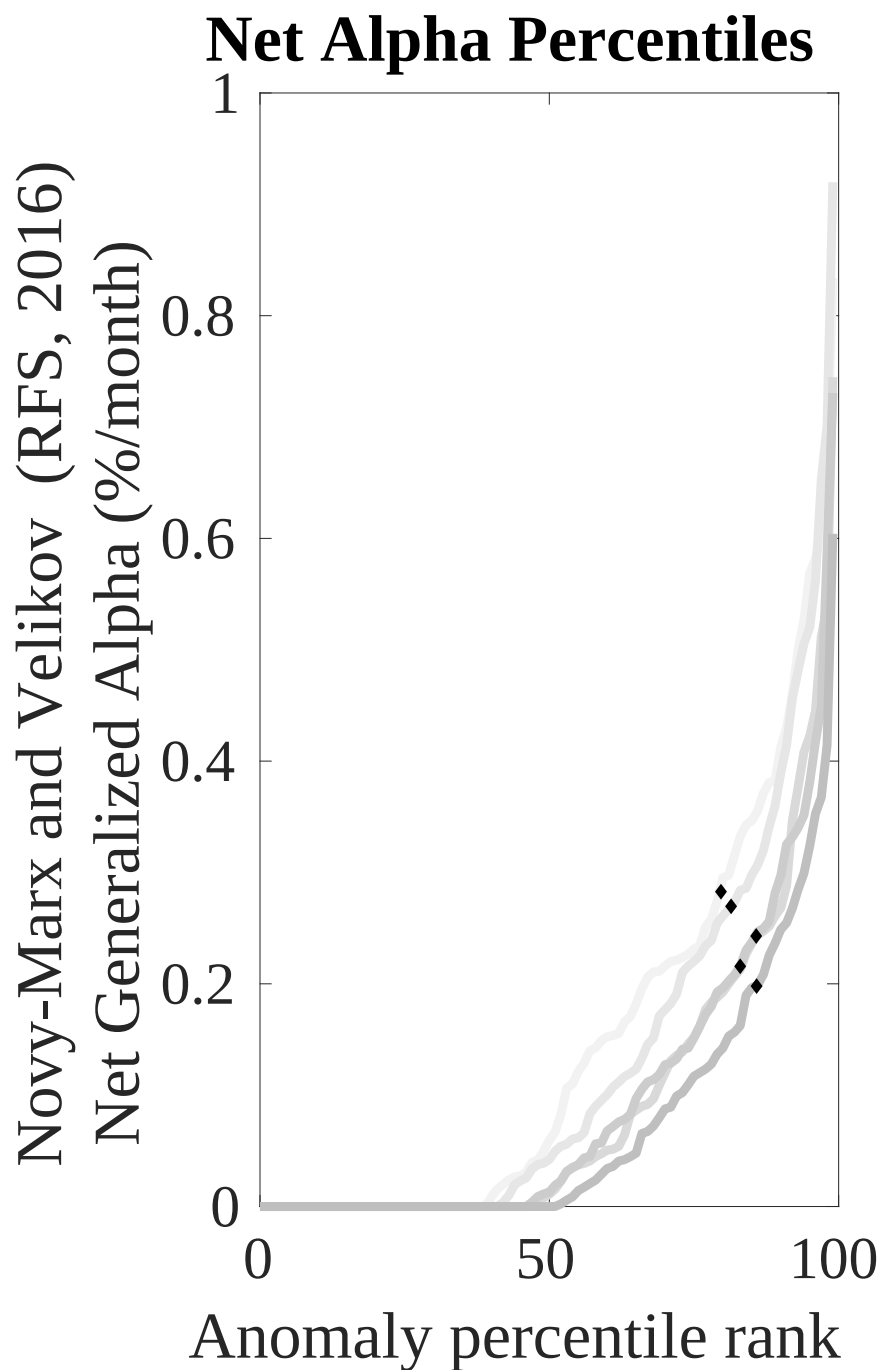
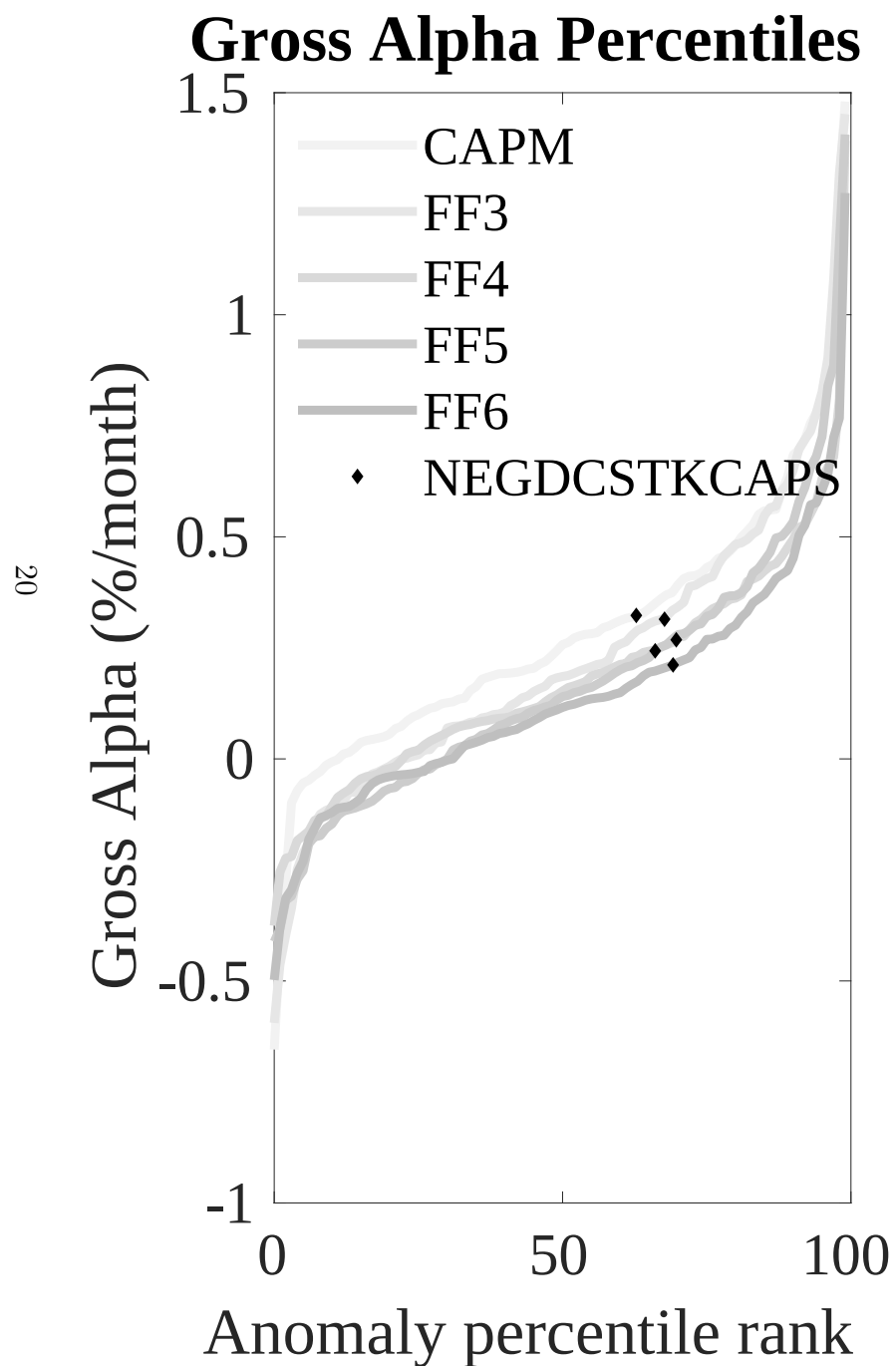


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CRD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



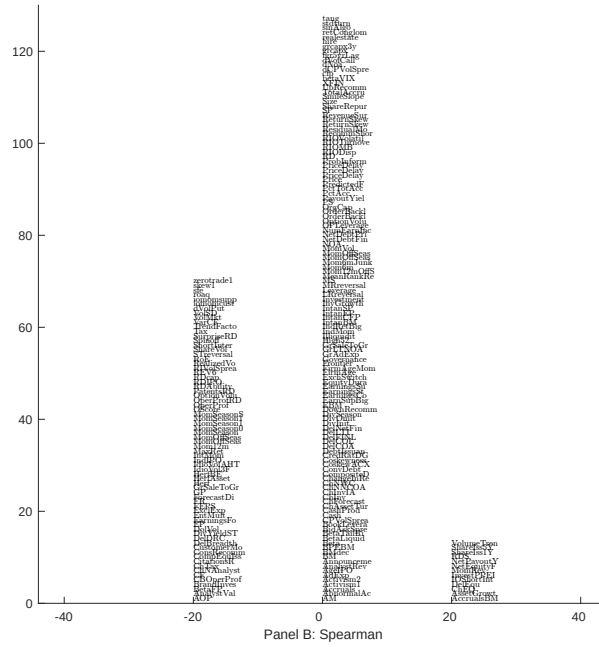
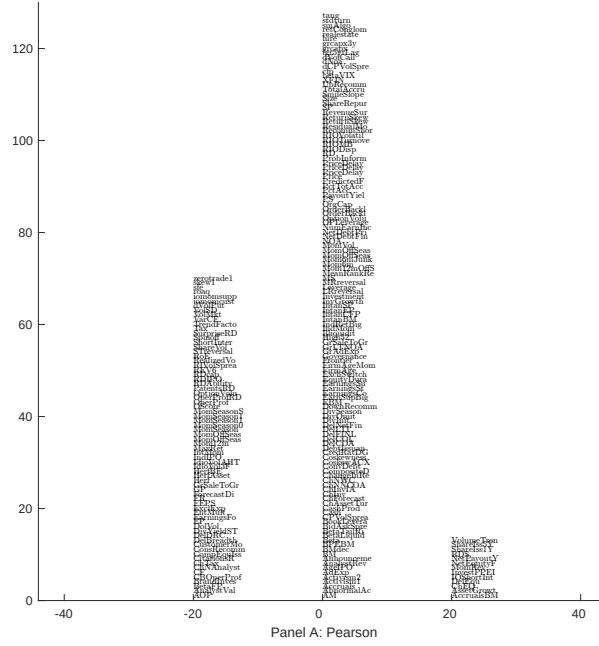


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with CRD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

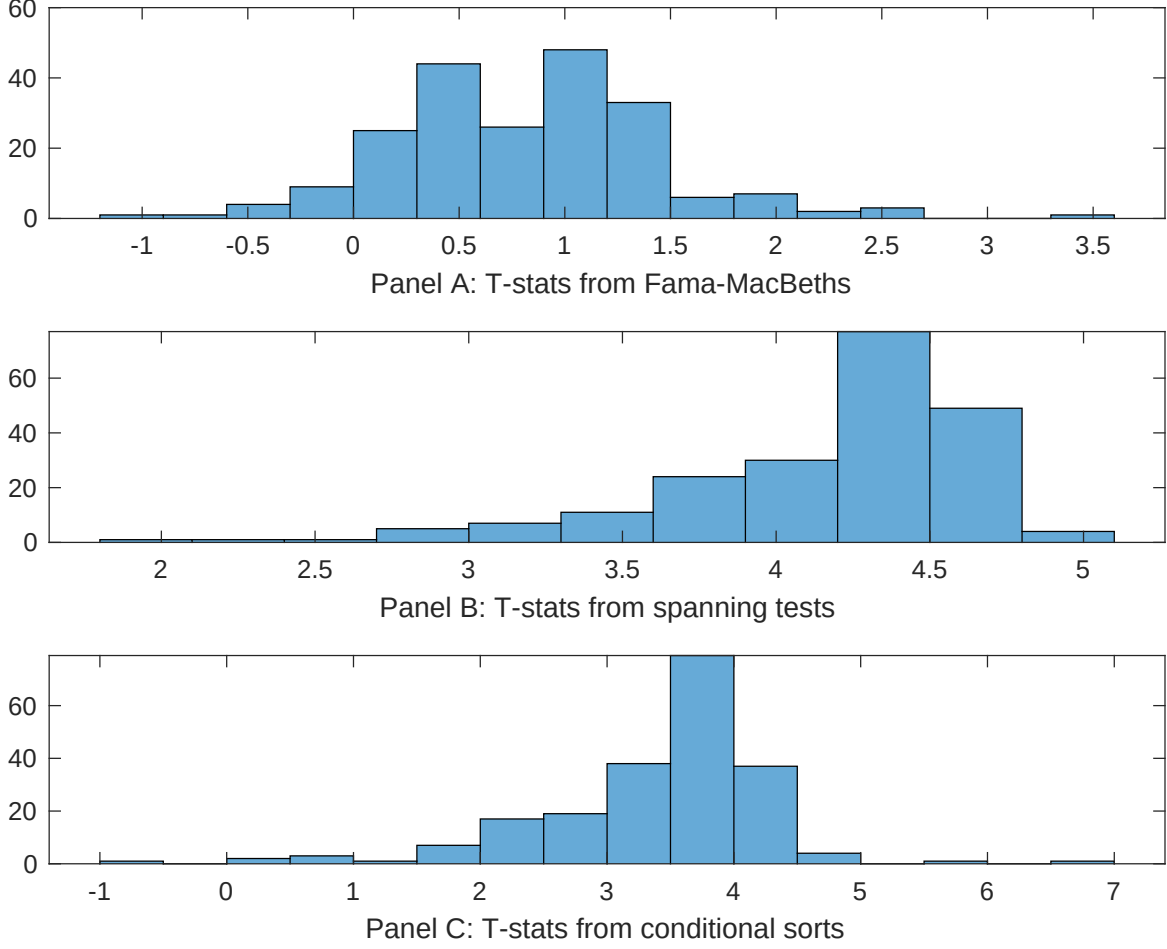


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CRD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRD}CRD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CRD. Stocks are finally grouped into five CRD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CRD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CRD}CRD_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.89]	0.17 [6.85]	0.13 [6.30]	0.12 [5.19]	0.12 [5.53]	0.13 [5.27]	0.12 [4.66]
CRD	0.19 [0.27]	0.99 [0.13]	-0.39 [-0.05]	0.38 [0.41]	0.35 [0.44]	0.62 [0.68]	0.22 [0.24]
Anomaly 1	0.15 [6.29]						0.79 [3.52]
Anomaly 2		0.43 [3.52]					-0.13 [-0.91]
Anomaly 3			0.28 [5.12]				0.27 [5.93]
Anomaly 4				0.23 [2.02]			0.13 [2.68]
Anomaly 5					0.12 [3.36]		0.13 [2.21]
Anomaly 6						0.18 [3.06]	-0.13 [-1.61]
# months	715	720	715	704	720	630	622
$\bar{R}^2(\%)$	0	0	0	1	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CRD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CRD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.19 [3.00]	0.20 [3.00]	0.13 [2.02]	0.24 [3.61]	0.21 [3.08]	0.22 [3.08]	0.16 [2.45]
Anomaly 1	32.53 [10.10]						15.11 [3.45]
Anomaly 2		21.20 [5.90]					20.38 [3.82]
Anomaly 3			31.99 [9.32]				21.63 [5.00]
Anomaly 4				15.75 [6.14]			12.14 [3.63]
Anomaly 5					6.82 [2.03]		-11.47 [-2.34]
Anomaly 6						11.19 [3.45]	-12.60 [-3.21]
mkt	-0.15 [-0.10]	-0.23 [-0.14]	0.49 [0.32]	1.51 [0.93]	-1.05 [-0.65]	1.34 [0.76]	1.48 [0.89]
smb	-0.29 [-0.13]	-2.25 [-0.98]	0.56 [0.25]	0.93 [0.40]	-1.63 [-0.70]	6.92 [2.47]	1.01 [0.38]
hml	-6.79 [-2.32]	-8.39 [-2.72]	-3.35 [-1.13]	-11.95 [-3.70]	-6.68 [-2.12]	-7.99 [-2.49]	-9.89 [-3.12]
rmw	-13.04 [-3.79]	5.60 [1.81]	-6.88 [-2.13]	-4.57 [-1.33]	5.22 [1.65]	3.30 [0.87]	-7.67 [-1.91]
cma	5.61 [1.19]	5.26 [0.92]	10.29 [2.22]	12.46 [2.52]	18.93 [3.32]	13.89 [2.67]	-8.01 [-1.35]
umd	3.13 [2.08]	4.27 [2.72]	2.53 [1.65]	5.88 [3.73]	4.82 [3.01]	4.37 [2.64]	2.98 [1.91]
# months	728	732	728	728	732	630	626
$\bar{R}^2(\%)$	19	12	18	12	8	8	22

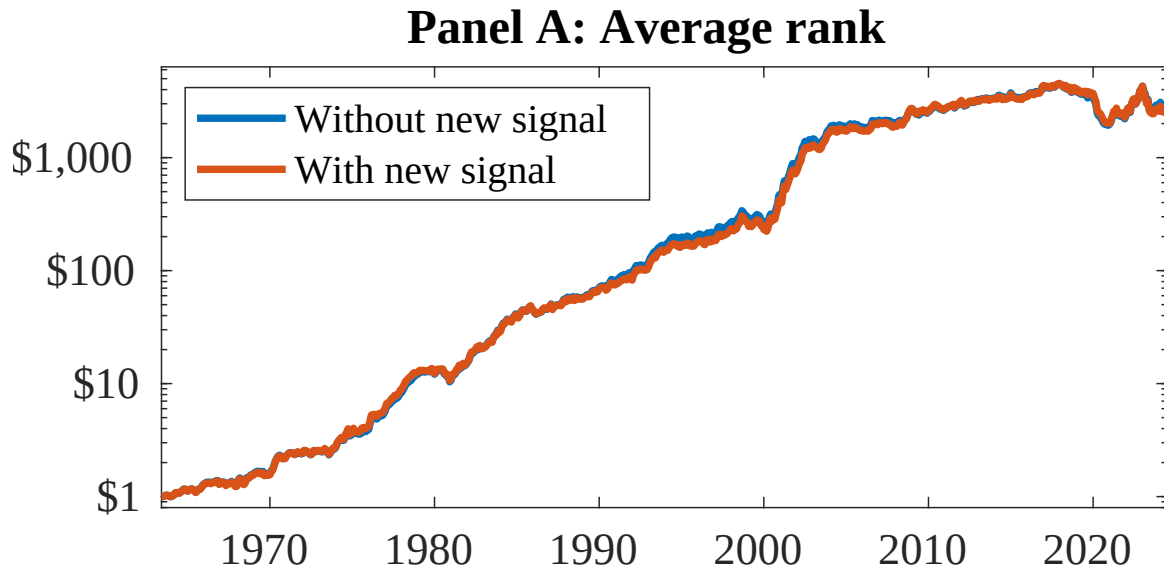


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CRD. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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